# Buddhadeb Mondal

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## Education

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| 2025 – present | **Postdoctoral researcher, Institute of Physics, Czech Academy of Sciences, Prague, Czech Republic** |
| 2019 – 2024 | **Ph.D., University of Siegen, Siegen, Germany**<br>Thesis: *Measurement of differential cross-sections of $t\bar{t}\gamma$ process with the ATLAS detector* |
| 2016 – 2018 | **M.Sc. Physics, Indian Institute of Technology Bombay, Mumbai, India**<br>Thesis: *Jet physics at the LHC* |
| 2013 – 2016 | **B.Sc. Physics, Jadavpur University, Kolkata, India** |

## Publications

1. ATLAS Collaboration, "Projected Sensitivity to Top-Quark Couplings in $t\bar{t}\gamma$ and $t\bar{t}Z$ Production at the HL-LHC with the ATLAS Experiment," 2025, ATL-PHYS-PUB-2025-015.
2. ATLAS Collaboration, "Measurements of inclusive and differential cross-sections of $t\bar{t}\gamma$ production in pp collisions at $\sqrt{s}$ = 13 tev with the atlas detector," *Journal of High Energy Physics*, vol. 2024, Oct. 2024. DOI: 10.1007/JHEP10(2024)191.
3. ATLAS Collaboration, "Measurement of the charge asymmetry in top-quark pair production in association with a photon with the ATLAS experiment," *Physics Letters B*, vol. 843, p. 137 848, 2023, ISSN: 0370-2693. DOI: https://doi.org/10.1016/j.physletb.2023.137848.

- Co-authored over 300 publications as part of ATLAS collaboration

## Skills

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| Language | Proficient in English |
| Programming language | C++, Python, Bash, Fotran, CMake |
| ML packages | PyTorch, TensorFlow, Scikit-learn |
| Data Analysis | ROOT, RooFit/RooStats, pandas, numpy, matplotlib |
| Simulation | PYTHIA8, MadGraph, DELPHES |
| Web Dev | HTML, CSS, JavaScript, Markdown |
| Development tool | CLion (IDE), Docker, Singularity, GNU Debugger, Valgrind, Intel VTune, |
| Misc. | Git, LaTeX |

## Projects

- **Calibration of ATLAS forward calorimeter cluster energy using Deep Neural Networks** *(Ongoing)* — with *Oldřich Kepka*
Many physics analyses, including vector boson scattering measurements, rely on signatures in the forward region of the ATLAS calorimeter. Because the ATLAS inner detector does not provide tracking coverage in this region, calorimeter clusters are the primary reconstructed objects available for energy measurements. Improving their energy calibration and pileup rejection could therefore benefit a broad range of physics analyses. In this project, we investigate machine-learning techniques to improve cluster energy reconstruction and pileup suppression beyond the performance of established methods.

- **Performance of Machine-Learning calibrated clusters in jet and hadronic recoil reconstruction** *(Ongoing)* — with *Oldřich Kepka*
Accurate reconstruction of the hadronic recoil is essential for precision measurements and searches involving missing transverse momentum. This project evaluates newly developed machine-learning-calibrated calorimeter clusters and investigates their impact on jet reconstruction, hadronic recoil resolution, and missing transverse momentum measurements.

- **Projection studies of top quark couplings at the HL-LHC** *(2024-2025) - Supervisor: Carmen Diez Pardos*
- Conducted sensitivity studies of top quark pair production with a photon and Z boson at the HL-LHC using Effective Field Theory interpretations.
- Extrapolated ATLAS Run 2 data (140 $fb^{-1}$) to high-luminosity scenarios (4–6 $ab^{-1}$) to derive projected limits on electroweak EFT operators.

- **Measurement of differential cross-section of $t\bar{t}\gamma$ process** *(2022-2024) - Supervisors: Carmen Diez Pardos, Ivor Fleck*
- Measured differential cross-sections across various observables characterizing the $t\bar{t}\gamma$ process, using proton-proton collision data from the ATLAS detector during Run 2.
- Prepared and published HEPData for this measurement. Available at `https://doi.org/10.17182/hepdata.146899`

- **Determination of backgrounds from mis-reconstructed electrons in $t\bar{t}\gamma$ topology** *(2020-2021) - Supervisors: Carmen Diez Pardos, Ivor Fleck*
- Studied backgrounds arising from electron mis-reconstruction as photons within the ATLAS detector.
- Estimated these backgrounds using data-driven techniques.

- **Determination of background from non-prompt lepton events in $t\bar{t}\gamma$ topology** *(2021-2022) - Supervisors: Carmen Diez Pardos, Ivor Fleck*
- Studied non-prompt lepton events in the $t\bar{t}\gamma$ topology, where leptons originate from b/c quark decays or mis-reconstruction.
- Estimated these backgrounds using data-driven techniques.

## Projects (continued)

- **ATLAS pixel DAQ development** *(2019-2020) - Supervisor: Oldrich Kepka*
  - Worked on development of the ATLAS pixel raw data analysis package for analyzing and validating data acquisition (DAQ) logic at the hardware and firmware level by decoding raw data from the detector.
  - Led a major upgrade of the package, including decoding information from the byte stream and designing a data structure for disk storage and further analysis.
  - Created an analysis framework for processing the decoded information.
  - Developed software to run computations on high-performance computing resources (Grid, HTCondor).
  - Acquired in-depth knowledge of the ATLAS pixel byte stream.

- **ATLAS pixel byte stream converter software** *(2023)*
  - This package is used for decoding ATLAS pixel byte stream, which is later utilized for physics object reconstruction.
  - Developed a feature which summarizes and logs errors detected during the byte stream decoding.

- **Jet physics at the LHC** *(2017-2018) - M.Sc. Project, TIFR, Mumbai - Supervisor: Manoranjan Guchait*
  - Conducted an in-depth study of jet properties within the LHC environment.
  - Focused on boosted jets, studied jet substructure techniques.
  - Evaluated performance of different top quark taggers in the boosted regime.

- **Top quark tagging using CNN** *(2018) M.Sc project, TIFR, Mumbai - supervisor: Manoranjan Guchait*:
  - It was an attempt to learn and replicate existing research on top quark tagging using jet images.
  - Exploited jet substructure techniques for top quark tagging.
  - Implemented convolutional neural network (CNN) based tagger using jet images.
  - Achieved performance comparable to the existing benchmarks.

- **Studying anisotropy in cosmic rays with the AMS-02 experiment** *(2017) - Internship Project, KIT, Germany - Supervisors: Fabian Bindel, Iris Gebauer*
  - Analyzed cosmic ray data from AMS-02 detector on ISS, focusing on Helium and Carbon events from first five years of data takings.

## Experiences

### Organizing workshops

- **ClusterML workshop @ FZU Prague** *(2026)* – Organized a one-week hackathon at FZU Prague focused on ongoing clusterML efforts.

### Software manager

- **Analysis software manager** *(2020-2021)* - Manager of the analysis software used by ATLAS top quark group.
- **Derivation software manager** *(2021-2022)* - Manager of the software used by ATLAS top quark group for producing Derived Analysis Object Data, a key data processing step for analysis.

### Teaching

- Delivered a comprehensive lecture on practical statistics for the master's lab course.
- Supervised an M.Sc. practical project titled "**Top Quark Physics at the LHC**".

## Experiences (continued)

- Acted as a **teaching assistant for the advanced particle physics course**, guiding students through solving exercises and complex concepts.
- Conducted **a tutorial session on ROOT** for master's students in a laboratory course.
- Conducted **a tutorial session on AnalysisTop framework**, an analysis framework used by the ATLAS top quark group, at
  - **ATLAS Germany meeting, Siegen, 2022**
  - **ATLAS top workshop, 2021**

### Leadership

- **ATLAS Germany local student representative** from the university

## Awards and Fellowships

- 2017 — **KHYS Grant**, Received internship grant from Karlsruhe House of Young Scientists, KIT, Germany for research work with the AMS-02 experiment.
- 2013-2018 — **INSPIRE Fellowship** - Department of Science and Technology, Government of India.
- 2013-2016 — **Central Sector Scheme for Scholarship (CSSS)** - Awarded by the West Bengal state government, India.

## Conference Talks

- **BOOST2026, Jagiellonian University, Krakow**
  Machine Learning Approaches to the Calibration of the Signals and their Classification as Pile-up in the ATLAS Calorimeters
- **QCD@LHC2025, Stony Brook University, New York**
  ATLAS Results on $t\bar{t}$ Cross sections and associated top quark production
- **DIS2023, Michigan State University, East Lansing**
  Associated top-quark production and measurement of production asymmetries with the ATLAS experiment, on behalf of ATLAS collaboration
- **ATLAS top workshop, 2023, Nikhef, Amsterdam**
  Status of ongoing ATLAS Run2 top+$\gamma$ measurements: $t\bar{t}\gamma$, $t\bar{t}\gamma\gamma$, $tq\gamma$
- **DPG spring meeting, 2023, Dresden**
  $t\bar{t}\gamma$ differential cross-section measurement with the ATLAS detector
- **Terascale 2022, Desy, Hamburg**
  $t\bar{t}\gamma$ differential cross-section measurement with the ATLAS detector
- **ATLAS Germany meeting, 2022, Siegen**
  $t\bar{t}\gamma$ differential cross-section measurement in single-lepton channel with the ATLAS detector
- **DPG spring meeting 2022, Heidelbeg, Online**
  $t\bar{t}\gamma$ differential cross-section measurement in dilepton channel with the ATLAS detector
- **DPG spring meeting 2021, Göttingen**
  Determination of background from mis-reconstructed electrons in $t\bar{t}\gamma$ topologies with the ATLAS detector